

E-band InP-on-Si Tunable Laser via Micro-Transfer Printing

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Abstract We demonstrate a widely tunable E-band InP-on-Si laser heterogeneously integrated using micro-transfer printing. The laser achieves a tuning range of 41 nm (1398–1439 nm), with a side-mode suppression ratio above 45 dB and 3 mW output power in the silicon waveguide. ©2026 The Author(s)

Introduction

Silicon photonics provides a CMOS-compatible platform for dense integration of low-loss components, but it lacks efficient light sources due to the indirect bandgap of silicon. Integrating III–V materials such as indium phosphide (InP) and gallium arsenide (GaAs) on silicon provides optical gain and enables on-chip laser sources, making heterogeneous integration a key technology for silicon photonics [1]. Over the past decade, various heterogeneous integration approaches, including wafer bonding, flip-chip integration, and micro-transfer printing, have enabled different integrated laser demonstrations [2–4]. In particular, tunable laser architectures based on external cavities with wavelength-selective feedback, such as distributed Bragg reflectors and ring resonators, have been widely demonstrated in both O-band and C-band [5–8].

Beyond the traditional O-band and C-band, silicon photonics also shows strong potential in the E-band (1360–1460 nm). This wavelength range is attractive for applications such as spectroscopy, environmental monitoring, and sensing [9]. Moreover, with the development of low-water-peak optical fibers, transmission loss in the E-band has been significantly reduced, making this wavelength range increasingly attractive for wavelength-division multiplexing (WDM) systems and optical communications. Recently, E-band lasers based on heterogeneous integration have been demonstrated, for example wafer-bonded InP-on-Si laser covering the wavelength range around 1350–1408 nm [10]. However, extension towards the upper E-band with wide tunability and high side-mode suppression ratio remains highly desirable for future integrated photonic systems.

In this work, we demonstrate a tunable InP-on-Si laser covering 1398–1439 nm, integrated on the IMEC silicon photonics platform using micro-transfer printing (μ TP) technology. Wide wavelength tuning is achieved using two silicon ring resonators with slightly different radii forming a Vernier mirror. The device can be continuously

tuned and exhibits a side-mode suppression ratio (SMSR) larger than 45 dB over the entire tuning range. The laser shows a threshold current of 70 mA and delivers up to 3 mW optical power in the silicon waveguide. This approach enables localized integration of III–V gain materials on 200 mm silicon-on-insulator (SOI) wafers, providing a scalable solution for heterogeneous integration without requiring modifications to either the III–V or CMOS foundry processes.

Design and fabrication

A schematic of the InP-on-Si tunable laser is shown in Fig. 1(a). The device is based on an extended-cavity geometry consisting of a wavelength selector on one side, an InP semiconductor optical amplifier (SOA) in the center, and a tunable reflector on the other side. Wavelength selection is achieved using a pair of thermally tunable micro-ring resonators (MRRs) with slightly different radii of 24 μ m and 22 μ m. Each ring resonator has a free spectral range (FSR) of approximately 3.9 nm. A loaded quality factor of around 8500 is measured for the MRRs with a 610-nm-wide waveguide and a 350-nm coupling gap. These MRRs form a Vernier filter, providing an extended effective FSR of around 25 nm compared to the individual resonators, enabling wide wavelength tuning. The other mirror consists of a tunable Mach–Zehnder interferometer (MZI) coupler combined with a loop mirror. The reflectivity can be controlled by adjusting the phase difference between the two MZI arms, allowing optimization of the cavity reflectivity for different operating wavelengths. In addition, a thermo-optic phase tuning section is included to adjust the cavity mode position. The InP SOA, grown and fabricated by III–V Lab, has a length of 1.2 mm and includes adiabatic tapers on both sides to ensure efficient coupling between the SOA and the underlying silicon waveguide. By modifying the composition of the epitaxial layers, the gain spectrum was shifted towards E-band range, with a photoluminescence peak centered at 1400 nm.

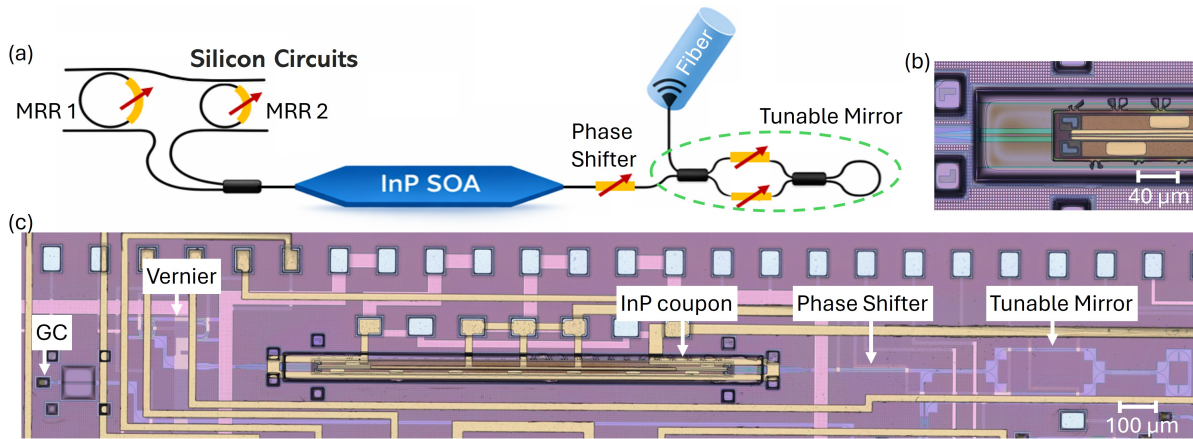


Fig. 1: (a) Schematic of the InP-on-Si tunable laser. (b) Transfer-printed InP coupon into a recess on the target silicon photonics chip. (c) Microscope image of the fabricated device. (MRR: micro-ring resonator, GC: grating coupler)

Heterogeneous integration

For the heterogeneous integration of the InP amplifier into the recess on the passive silicon chip, the micro-transfer printing technique is used [11]. Prior to printing, a 100 nm DVS-BCB layer is spray-coated on the silicon target chip as an adhesive layer, followed by a soft bake at 150 °C for 15 min. The μ TP process is performed using an X-Celeprint μ TP-100 lab-scale tool. The process relies on the kinetically controlled adhesion of an elastomeric stamp to pick up pre-processed InP coupons from the source wafer and print them onto the target silicon photonics wafer. Fig. 1(b) shows the transfer-printed coupons with a lateral misalignment of less than 500 nm with respect to the underlying silicon waveguide. Compared to wafer bonding techniques, micro-transfer printing allows integration of III–V coupons into pre-defined recesses without requiring any modification to the CMOS fabrication process. The integration approach allows independent fabrication and testing of both the silicon photonic circuits and the III–V devices in their respective foundries, which improves the overall integration yield. Moreover, this approach enables massively parallel integration, paving the way for wafer-scale manufacturing. After transfer printing, an additional metallization step is performed to redistribute the electrical contacts for probing. Fig. 1(c) shows the optical microscope image of the fabricated laser.

Characterization

The InP-on-Si tunable laser was electrically driven using DC probes on a temperature-controlled stage maintained at 18 °C. Multiple Keithley 2400 source meters were used to bias the III–V gain section and the thermal phase tuning elements, including MRRs, phase shifter, and tunable loop mirror. Light was extracted using a single-mode fiber (SMF) through a grating coupler (GC). A reference silicon waveguide with identical grating couplers on both sides was used to calibrate the cou-

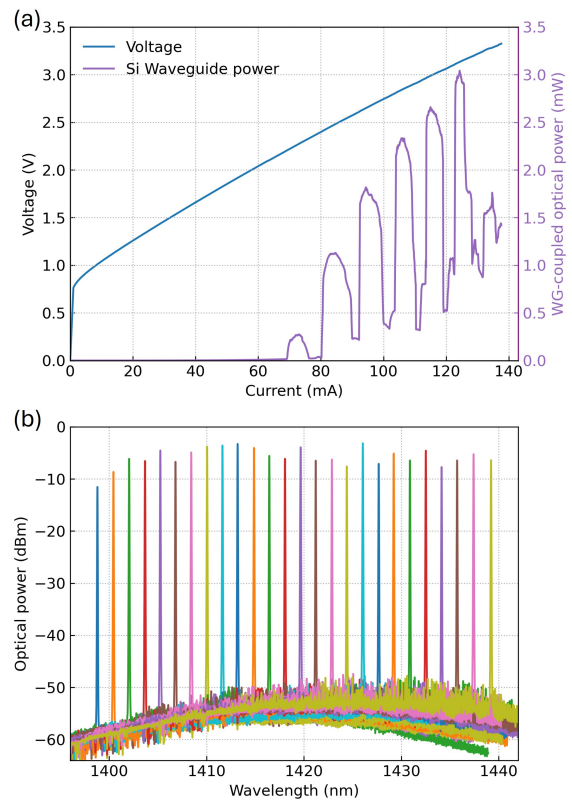


Fig. 2: (a) Light-current-voltage curve of the tunable laser. (b) Optical spectra under different operation wavelengths.

pling loss. Fig. 2(a) shows a typical measured light-current-voltage (LIV) curve. The threshold current is approximately 70 mA, and the calibrated output power in the silicon waveguide reaches 3 mW at a bias current of 124 mA. During the LIV measurement, only the gain current was swept while all heater biases were kept constant. The periodic oscillation of the output power results from the thermally-induced phase changes in the gain section, which shift the cavity mode positions. High output power occurs when the cavity modes align with the MRR resonances, while lower output power occurs when the cavity modes are detuned from the resonances.

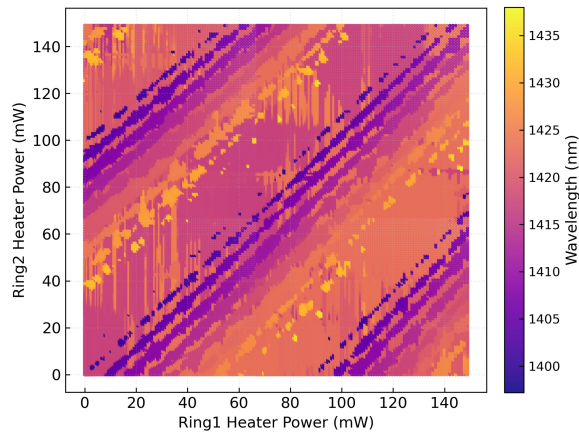


Fig. 3: 2D map of the lasing wavelength as a function of the two ring heater powers.

The optical spectra at different operating wavelengths were measured using an optical spectrum analyzer with a resolution of 30 pm, as shown in Fig. 2(b). The laser exhibits a wide tuning from 1398 to 1439 nm, covering a total tuning range of 41 nm, with SMSR larger than 45 dB. As expected, the wavelength tuning breaks the limitation of Vernier FSR and extends to two periods [12].

Fig. 3 shows the wavelength tuning map. For this measurement, both ring heaters were iteratively swept across a full Vernier free spectral range. The gain current was fixed at 100 mA, while the phase tuning section and tunable loop reflector were kept at constant bias. Optical spectra were recorded at each step. The dominant lasing wavelength was extracted and plotted as a function of the two ring heater powers. The periodic pattern observed in the tuning map confirms the Vernier tuning mechanism between the two ring resonators.

Conclusion

We demonstrated an E-band InP-on-Si tunable laser heterogeneously integrated on the IMEC 200 mm silicon photonics platform using micro-transfer printing. The laser delivers up to 3 mW output power in the waveguide and covers a tuning range from 1398 to 1439 nm. The use of pre-fabricated III-V gain enables independent fabrication and quality control of the III-V devices and silicon PICs in their respective foundries, improving the yield and scalability of heterogeneous integration. Future work will focus on integrating SOAs with different gains on the same chip to realize ultra-wideband tunable laser arrays. Such laser arrays could cover the entire low-loss fiber telecommunication window. Additional amplifiers could also be integrated in the same way to further increase the on-chip optical power [13].

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